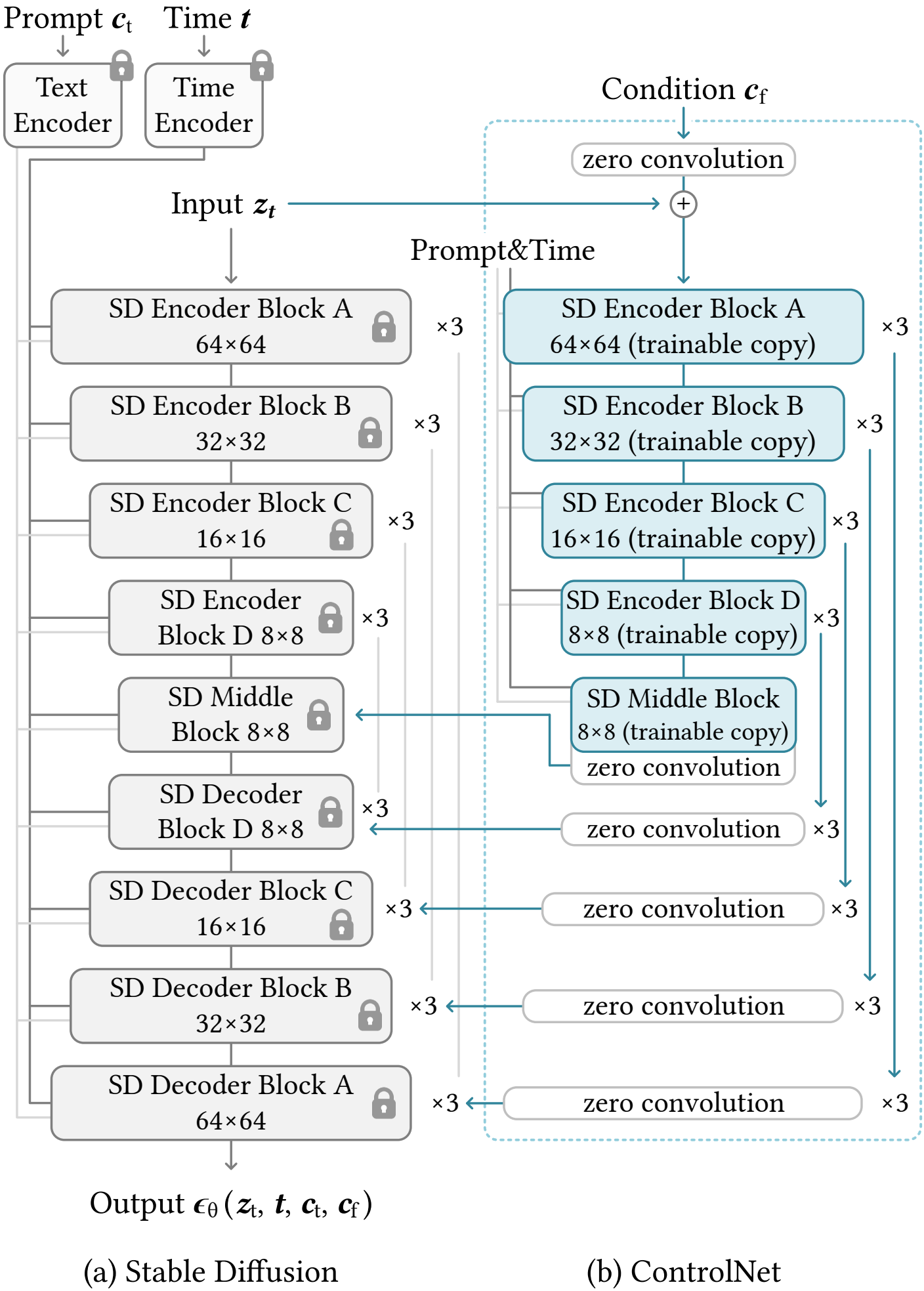

Prompt c_t
Time t
Text Encoder
Time Encoder
Condition c_f
zero convolution
Input z_t
Prompt&Time
SD Encoder Block A 64×64
×3
SD Encoder Block A 64×64 (trainable copy)
×3
SD Encoder Block B 32×32
×3
SD Encoder Block B 32×32 (trainable copy)
×3
SD Encoder Block C 16×16
×3
SD Encoder Block C 16×16 (trainable copy)
×3
SD Encoder Block D 8×8
×3
SD Encoder Block D 8×8 (trainable copy)
×3
SD Middle Block 8×8
SD Middle Block 8×8 (trainable copy)
zero convolution
SD Decoder Block D 8×8
×3
zero convolution
×3
SD Decoder Block C 16×16
×3
zero convolution
×3
SD Decoder Block B 32×32
×3
zero convolution
×3
SD Decoder Block A 64×64
×3
zero convolution
×3
Output $\epsilon_\theta(z_t, t, c_t, c_f)$
(a) Stable Diffusion
(b) ControlNet